Bakery Sales Prediction

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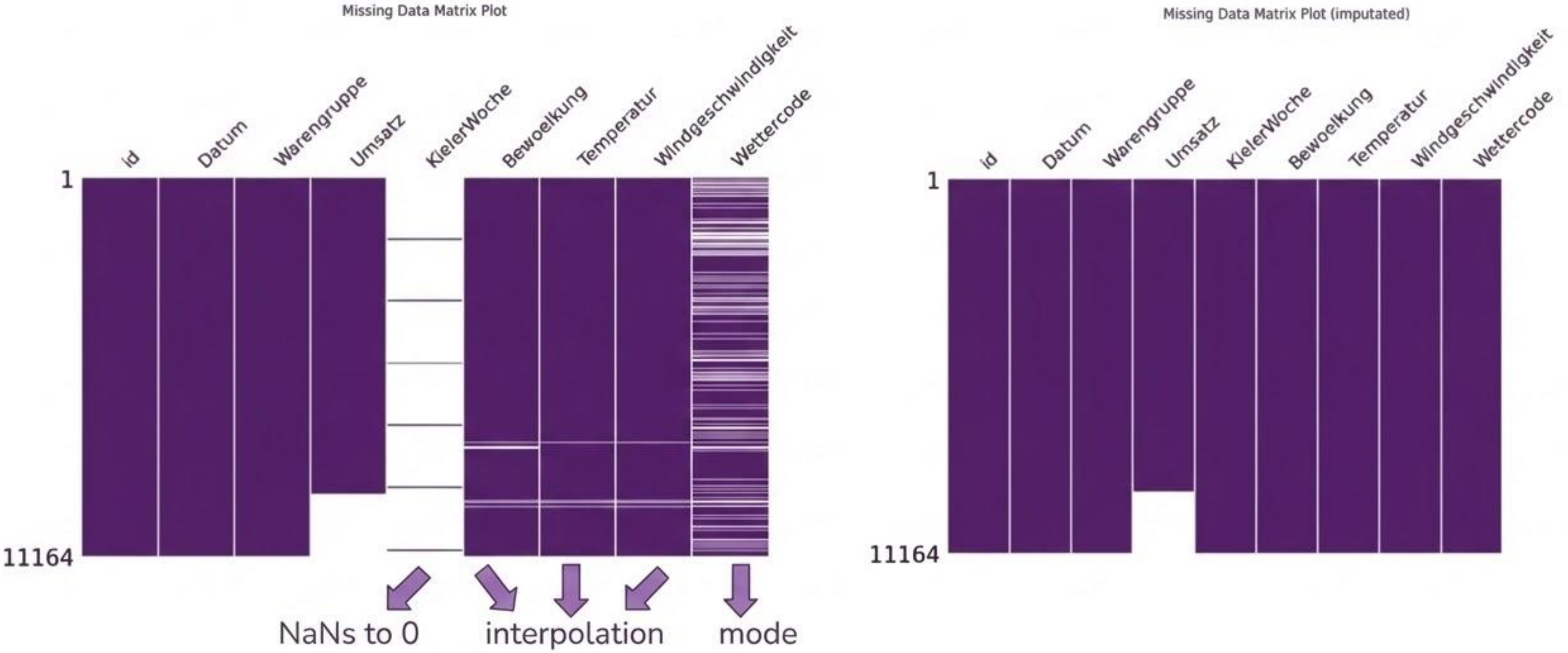
Introduction to Data Science and Machine Learning - W25/26 - Group 14


Holiday
Weather
Time
Data Exploration and Cleaning
Feature Engineering
Baseline Model - Linear Regression
Decision Trees - RF and XGBoost
Neural Network

Data Cleaning & Imputation

Missing Value Imputation



Data Imputation Analysis

Bewölkung

	Before Imputation		After Imputation
Mean:	4.92	~	4.86
Var:	52.03		12.16

-76.61%

Temperatur

	Before Imputation		After Imputation
Mean:	12.19	~	12.05
Var:	17.08		6.50

-61.90%

Windgeschwindigkeit

	Before Imputation		After Imputation
Mean:	11.01	~	12.17
Var:	7.01		3.22

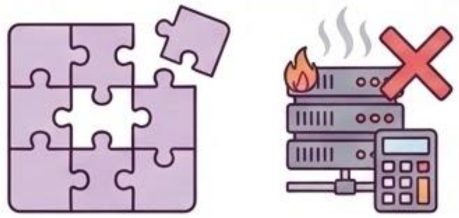
-54.14%

→ very similar means

→ lower variance
= more stability!

Why KNN was not used:

Few Missing Data



No need for □ comp. cost

No Temporal Awareness

Outliers & Noise Risk

Interpolation and **Mode** includes these aspects

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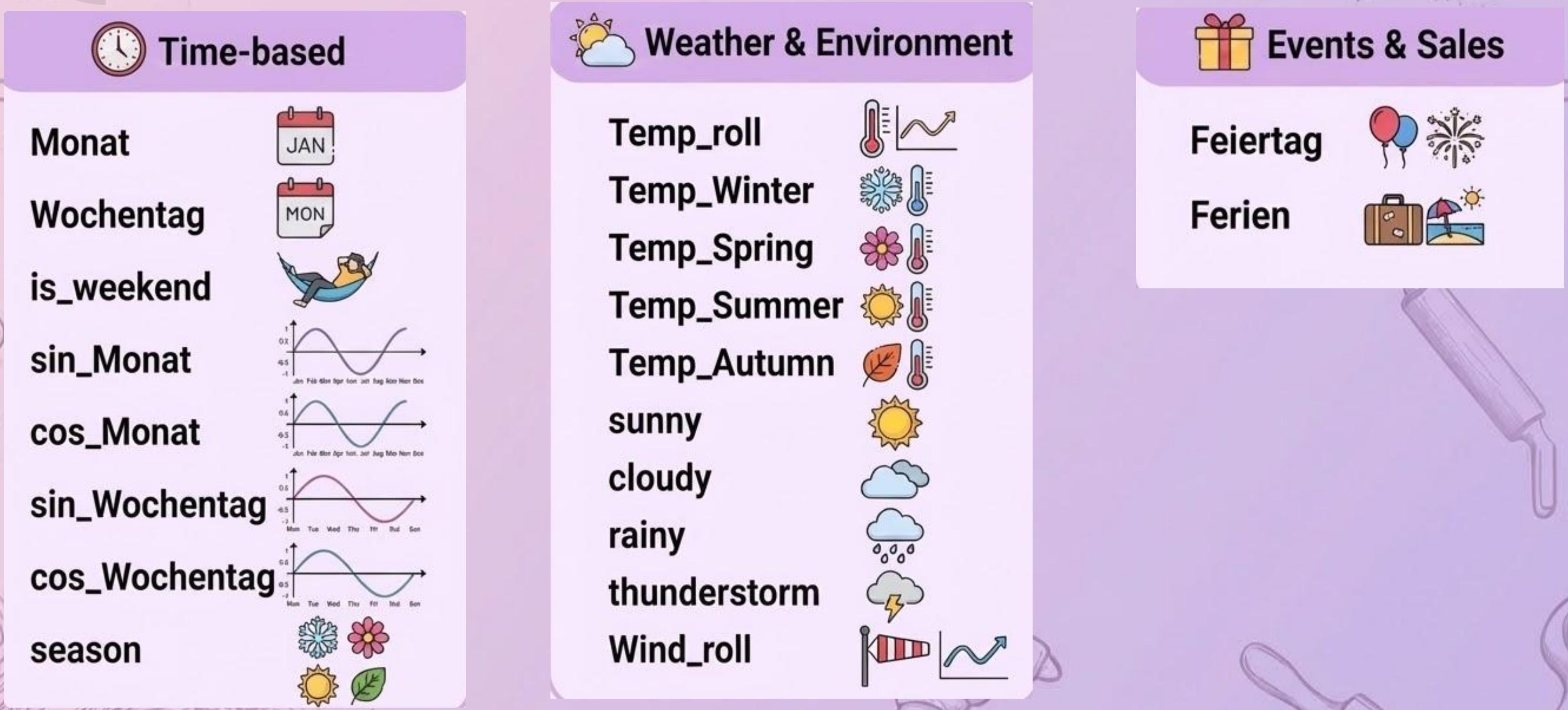
Feature Engineering



Worst Fail

Not one-hot-encoding Weekdays

Temporal not one-hot-encoded:
→ better model performance

One-hot-encoded:
→ worse model performance

Why?

Monday→Sunday:
1→7

Monday→Sunday:
1/0

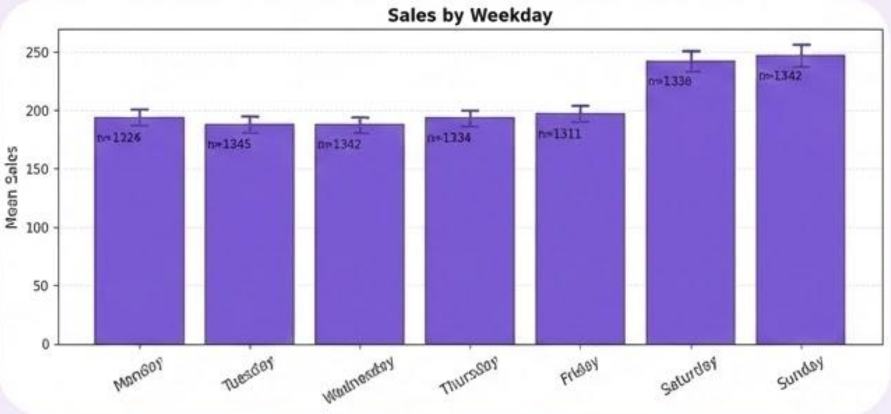
Weekend:
more weight

Conclusion: Weekend very important variable

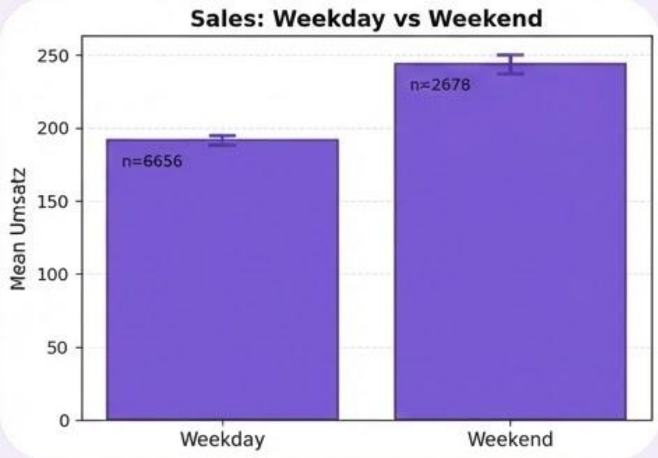
Feature Engineering

Visuals

Sales by Weekday



Sales: Weekday vs Weekend

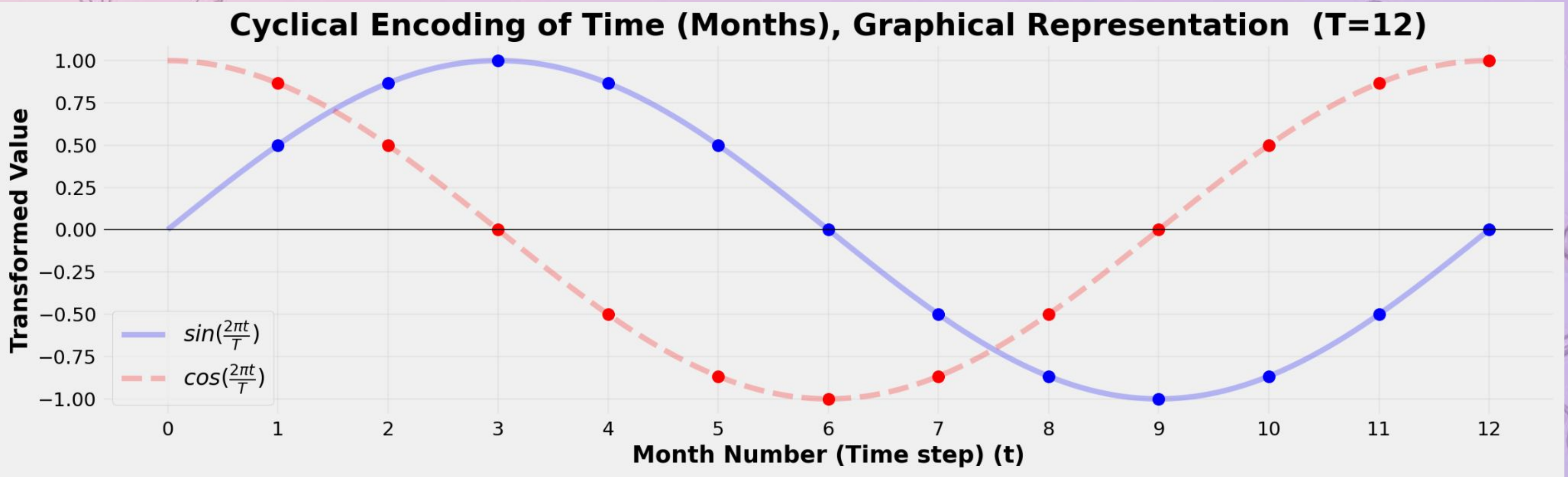


Time Based Feature Engineering - Fourier Terms

Capture Seasonality / Cyclic Effect

Two Fourier Terms for each **Month & Weekday**

Best Improvement

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Multiple Linear Regression

Model Choice

Why?

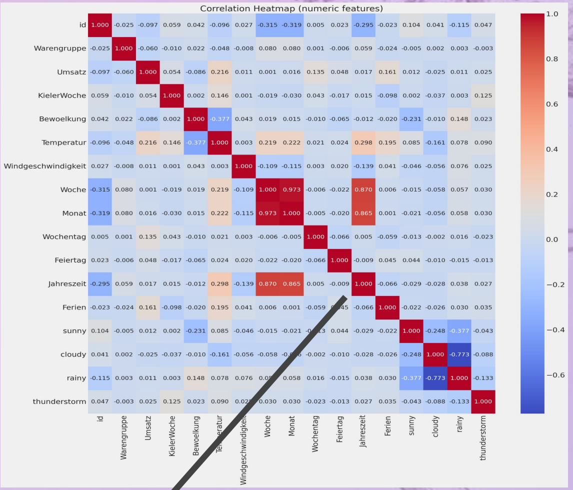
- Classical approach
- Easy and fast to run
- Low computation needs

But is it good for Time Series problems?

- Interesting results through **Clever Feature Engineering**

Multiple Linear Regression

Feature Selection



How we approached the issue?

- More time-based feature engineering
- Interaction among variables

Overview of the Correlation Matrix shows the way...

Temperatur

Wochentag

Ferien

Umsatz	-0.097	-0.060	1.000	0.054	-0.086	0.216	0.011	0.001	0.016	0.135	0.048	0.017	0.161	0.012	-0.025	0.011	0.025

KielerWoche

Bewölkung

Feiertag

Multiple Linear Regression

Category differences

External Factors

Temperature, Special Days, Kieler Woche.

Temporal Seasonality

Global seasonality patterns for specific weeks & months.

- Multiple product categories
- Distinct sales patterns

Multiple Linear Regression

Interaction among variables

$$y \approx \text{Global Intercept} + (\text{Product Groups} \times \text{Interaction Variables}) + (\text{Coefficients} \times \text{Global seasonality Variables})$$

```
formula = (
    "Umsatz ~ C(Warengruppe) * (KielerWoche + is_weekend + Temperatur + Ferien + Feiertag) "
    "+ sin_Monat + cos_Monat + sin_Wochentag + cos_Wochentag"
)

# Fit Model
model = smf.ols(formula=formula, data=train_df).fit()

# Print Summary to check R-squared and Adj. R-squared
print(model.summary())
```

	Linear Regression
Adj R^2 Val	0.813
MAPE Val	23.66
MAPE Public	21.52
MAPE Private	21.10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Tree Models

Best parameters

Random Forest:

- **n estimators: 600**
- **max depth: 20**
- min samples split: 20
- min samples leaf: 2
- max features: 1.0
- bootstrap: True

Deep trees constrained by **conservative split criteria**

XGBoost:

- **n_estimators: 1000**
- **max_depth: 7**
- subsample: 1.0
- reg lambda: 5.0
- reg alpha: 0.0001
- min child weight: 10
- learning rate: 0.01
- gamma: 0.0
- col sample by tree: 0.8

Slow learning with **strong regularization**


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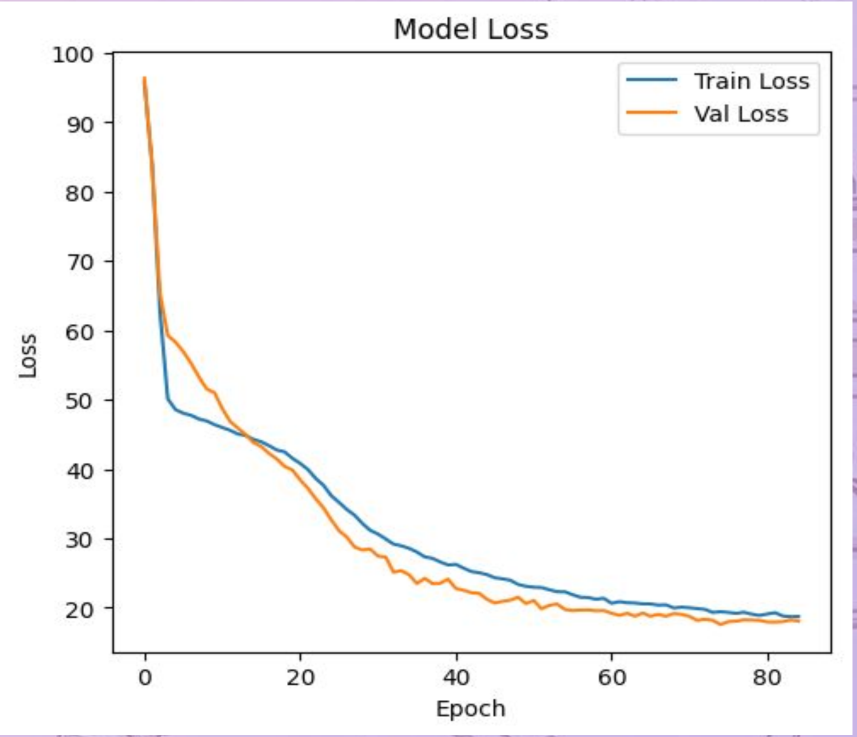
Neural Network Optimization

```
model = Sequential([
    InputLayer(shape=(X_train_s.shape[1],)),
    Dense(256, activation="relu"),
    BatchNormalization(),
    Dropout(0.2),

    Dense(128, activation="relu"),
    BatchNormalization(),
    Dropout(0.2),

    Dense(64, activation="relu"),
    Dense(1)
])
```



Neural Network Optimization

	3 dense layers
MAPE Val	18.14
MAPE Public	20.723
MAPE Private	19.880

Cat	MAPE %
1	19.06
2	**12.42**
3	17.98
4	22.39
5	15.03
6	**42.45**

Models Comparison

	Linear Regression	Random Forest	XGBoost	Neural Network
R^2 Val	0.814	0.818	**0.83048**	**0.83059**
MAPE Val	23.66	20.84	20.02	**18.14**
MAPE Public	21.52	**18.38**	19.07	20.72
MAPE Private	21.10	17.29	**17.03**	19.88
Best Cat MAPE (Cat 2)	18.56	16.29	15.64	**12.42**
Worst Cat MAPE (Cat 6)	65.35	**38.15**	38.28	42.45

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Questions?

Thank you!